AIST4010: Foundation of applied deep learning

Assignment 1

Out: Sep 24, 2026

Due: Oct 14, 2026 (11:59pm)

Note:

- Be sure to include your name and student ID in the cover page.
- You can either use an electronic version (strongly preferred) or the hand-written version. However, if you use the hand-written version, please scan it to one file.
- Please submit assignments as .pdf format and codes as .py or .ipynb through the blackboard before the deadline. That is, we will only accept assignments from the blackboard.
- Submitted work must be your own.

1. Please prove the following equation. (10 pts)

$$\frac{d}{dx} sigmoid(x) = sigmoid(x)\big(1 - sigmoid(x)\big).$$

2. Please prove that multi-layer linear neural network is equivalent to single layer neural network. (10 pts)

3. Given a dataset that contains ten data samples, each of which is represented by one feature. The feature values and the corresponding labels are given in the table below. Assume you are asked to use threshold-based classifiers, considering label 1 as the positive class and 0 as the negative class, please draw the ROC curve for this dataset with respect to different values of the threshold and calculate the AUC. (Hint: no need to smooth the ROC curve) (10 pts)

Sample index	Feature values	Label
S1	0.7	0
S2	0.7	1
S3	0.9	1
S4	0.7	1
S5	0.6	1
S6	0.6	0
S7	0.1	1
S8	0.4	0
S9	0.3	1
S10	0.3	0

4. Let's implement logistic regression using different methods. Before we use PyTorch, let us build everything from scratch. That is, you are only allowed to use NumPy to perform basic calculation and optimization.
   We provide a sample python code file including the assignment pipeline, please follow the instructions and write the functions to finish this assignment. Remember to keep the seed not changed to make sure the result can be the same.
   a. Data generation: please generate a linearly separatable dataset with 100 features and 2 classes, each class having 1000 points. Please add some Gaussian noise into the data to make it not necessarily to be linearly separatable. Please also generate a testing dataset with 100 points. (4 pts)
   b. Please implement logistic regression only using NumPy and run GD to find your function to separate the data. Please record the train/test loss history and the parameter W & b, and calculate the accuracy. (5 pts)
   c. Please run SGD to find your function to separate the data. Please record the train/test loss history, the parameter W & b, and calculate the accuracy. (3 pts)
   d. Please plot the train and test loss curves during training using the loss history. Save the figures and insert into your report. (3 pts)
   e. Please use PyTorch framework to resolve the problem without for loop. Also record the train/test loss history and the parameter W & b, calculate the accuracy, and plot the train/test loss curves. (5 pts)

5. In this question, you can use any framework and package. Please apply logistic regression, SVM, fully-connect neural networks, and CNN (if you can) on the Iris dataset to perform classification. Which one gives you the best performance? (6 pts) Why does not deep neural network show its power on this dataset? (4 pts)

6. Please summarize the below papers, using the following format. (5 * 8 pts)

   Motivation: Why did the authors want to do the work?
   Novelty: How is their method different from the previous works?
   Intuition: Why would their method work and be better than the previous methods intuitively?
   Implementation: The link to the code.

   Papers:
   ImageNet Classification with Deep Convolutional Neural Networks
   Deep Residual Learning for Image Recognition
   U-Net: Convolutional Networks for Biomedical Image Segmentation
   A Simple Framework for Contrastive Learning of Visual Representations
   Masked Autoencoders Are Scalable Vision Learners